Poker Theory & Bot Design: Lecture 3

Hand Selection, Position & the Lookup Bot

Poker Theory and Design of Poker Bots Reading Group

February 24, 2026

Meeting 3 Overview

1. Hand Strength
2. Tight vs. Loose Play & Range Analysis
3. Positional Advantage
4. Building the Lookup Bot
5. Bluffing & the 15% Surprise
6. Next Time: Bluffing Theory & the Road to GTO

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Let's Do Some Combinatorics

A standard deck: 13 ranks $\times$ 4 suits $=$ 52 cards. Each player receives 2 private ("hole") cards, unordered.

$$|\Omega| = \binom{52}{2} = \mathbf{1326} \quad \text{starting hands}$$

Suit symmetry (preflop): No community cards have been dealt, so suits are interchangeable. K♠A♡ $\equiv$ K♡A♢ strategically. This collapses to **169** canonical hand classes:

$$\begin{aligned} \text{Pairs} &: \binom{13}{1}\binom{4}{2} = 13 \times 6 = 78 \\ \text{Suited} &: \binom{13}{2} \times 4 = 312 \\ \text{Offsuit} &: \binom{13}{2} \times 12 = 936 \end{aligned}$$

Total: $78 + 312 + 936 = 1326$. Classes: $13 + 78 + 78 = \mathbf{169}$.

Hand Weights: What They Mean

Definition. The *weight* $w(h)$ of canonical hand h is the number of distinct card-pair deals in that class:

$$w(h) = \left|\{(c_1, c_2) \in \Omega : (c_1, c_2) \sim h\}\right|$$

Examples:

- $w(\mathrm{AA}) = \binom{4}{2} = 6$ (choose 2 from 4 aces: A♡A♢, A♡A♣, A♡A♠, A♢A♣, A♢A♠, A♣A♠)

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- $w(\mathrm{AKs}) = 4$ (one per suit: A♡K♡, A♢K♢, A♣K♣, A♠K♠)

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- $w(\text{AKs}) = 4$ (one per suit: A♡K♡, A♢K♢, A♣K♣, A♠K♠)
- $w(\text{AKo}) = 4 \times 4 - 4 = 12$ (all combos minus suited)

Suited, Connected, and Drawing Hands

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- Enables a *flush*

Connected (connectors): Adjacent ranks (differ by 1).

- Example: 9♢8♢
- Enables a *straight*

Hand Equity: Formal Definition

Definition (Equity). Let h_1, h_2 be two starting hands. B is drawn uniformly over all boards not conflicting with h_1, h_2.

$$\mathrm{Eq}(h_1 \mid h_2) = \Pr[\mathrm{rank}(h_1, B) > \mathrm{rank}(h_2, B)] + \frac{1}{2}\Pr[\mathrm{rank}(h_1, B) = \mathrm{rank}(h_2, B)]$$

The $\frac{1}{2}$ handles splits (ties). We can come up with a more complicated formula for more players using inclusion-exclusion principal.

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Selected equities (R code):

h_1	h_2	$\mathrm{Eq}(h_1)$
`AA`	`KK`	81.8%
`AA`	`72o`	95.7%
`AKs`	`72o`	73.4%

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Range equity. In practice we don't know opponent's exact hand. We compute equity against their *range* $\mathcal{R}$:

$$\mathrm{Eq}(h_1 \mid \mathcal{R}) = \sum_{h_2} \mathcal{R}(h_2) \cdot \mathrm{Eq}(h_1 \mid h_2)$$

Equity Is Not EV!

Equity: Who wins if we were to move to showdown immediately.
Expected Value (EV) of a call accounts for:

- **Pot odds:** you risk C to win a pot of P

Figuring out the EV is quite hard and computationally infeasible. We are trying to estimate it at varying levels of complexity. The simplest and most straightforward one being EV.

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- **Position:** acting last gives you information your opponent doesn't have

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Pot Odds and the Call Threshold

Recall. Given Pot $= P$, call costs C:

$$\text{Pot Odds} = \frac{C}{P + C}$$

That is the minimum equity needed to break even on a call.
If we use the same idea as the one from last time, we can have a decision rule as

$$\mathrm{Eq}(h_1 \mid \hat{h}_2) \geq \frac{C}{P + C}$$

where $\hat{h}_2$ is the opponent's range

Why Are Some Hands Strong?

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1. Raw equity dominance.

$$\mathrm{Eq}(\mathrm{AA} \mid h) \geq 0.78 \quad \forall\, h \neq \mathrm{AA}$$

2. Dominated kicker. Both players flop an Ace; whoever has the higher second card wins.

$$\mathrm{Eq}(\mathrm{A2o} \mid \mathrm{AKo}) \approx 28\%$$

Playing A2 often means paying to lose to AK.

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3. Drawing potential.

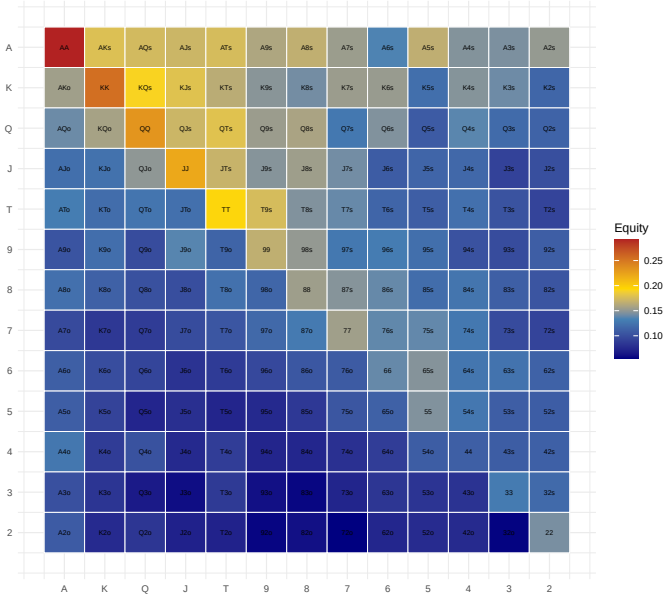
$$\Pr[\text{flush by river} \mid \text{suited}] \approx 5.9\%$$
$$\Pr[\text{straight} \mid \text{connectors}] \approx 4.6\%$$

High suited connectors (K♡Q♡) have both benefits.

4. Why 72o is worst.

7 and 2 cannot form any straight.
Offsuit means no flush draw.
$\mathrm{Eq}(72o \mid \text{random}) \approx 32.2\%$.

9−Handed Preflop All−In Equity (Monte Carlo)



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VPIP: Voluntarily Put $ In Pot

Definition. VPIP is the fraction of hands a player *voluntarily* enters the pot with (i.e. does not fold preflop, excluding the forced blind posts):

$$\mathrm{VPIP} = \frac{\sum_{h:\,\pi(h)\neq\text{fold}} w(h)}{\sum_h w(h)} \times 100\%$$

The $w(h)$ weighting is essential: playing KKo counts 12× as often as you are dealt it compared to playing it once.

Style	VPIP	Characteristic
Nit	$< 15\%$	Only strongest premiums
Tight	15–22%	**GTO-approximate region**
Loose	23–35%	Many speculative hands
Maniac	$> 35\%$	Plays most hands

Range Analysis: Thinking in Distributions

What is a range? Since we cannot see opponent's hole cards, we maintain a probability distribution over possible hands:

$$\mathcal{R}(h) = \Pr[h_{\text{opp}} = h \mid \text{observed actions}]$$

Initialised as $\mathcal{R}(h) = w(h)/1326$ (uniform prior), then updated as actions are observed.

Bayesian update. Let A_t be the action taken at time t and $\sigma^*(A_t \mid h)$ be the GTO probability of taking action A_t with hand h:

$$\mathcal{R}(h \mid A_1, \ldots, A_t) = \frac{\mathcal{R}(h) \prod_i \sigma^*(A_i \mid h)}{\sum_{h'} \mathcal{R}(h') \prod_i \sigma^*(A_i \mid h')}$$

Example. If someone raises from UTG (first to act, 9-handed), we are expecting them to have some of the best possible hands in the game and by no means have 72o. So we can say $\mathcal{R}(72\text{o}) approx 0$.

We must train our bots to play against this distribution, not the initial $\mathcal{R}(h) = \frac{w(h)}{1326}$.

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UTG and Other Positions: What They Mean

Positions in a 9-handed game (clockwise from dealer button):

- **BTN** (Button): dealer; acts last postflop — best seat

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- **MP** Middle Positions
- **CO** (Cutoff): one before BTN; second-best seat

Position Edge

Definition. Player i is *in position* (IP) if i acts *after* all opponents on every postflop street. Acting first = *out of position* (OOP).

The IP player observes the OOP check/bet before responding. Let $A_{\text{OOP}} \in \{\text{check}, \text{bet}\}$ carry $H(A_{\text{OOP}}) > 0$ bits. IP's strategy can condition on this signal; OOP cannot. The IP player's expected utility weakly dominates OOP's for the same hand.

GTO solver data (6-max, 100bb deep):

Position	Meaning	EV (bb/hand)	VPIP
BTN	Button, last postflop	$+0.42$	47%
CO	Cutoff	$+0.24$	32%
UTG + 1	Hijack	$+0.11$	27%
UTG	Lojack	-0.01	22%
BB	Big Blind (forced)	-0.12	—
SB	Small Blind (forced)	-0.27	—

Position VPIP: Why We Play More Hands Late

We play *wider* ranges from later positions:

$$\text{UTG range} \approx 12\text{–}15\% \qquad \text{CO range} \approx 25\text{–}30\% \qquad \text{BTN range} \approx 40\text{–}50\%$$

Why?

- IP, you can see the OOP player check-fold and take the pot cheaply

Bot design implication. Strategy must be $\pi(h, \text{pos})$, not just $\pi(h)$. The VPIP of our LookupBot should vary by seat.

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- IP, you can control pot size on draws (call small bets, fold big ones)
- The information advantage more than compensates for weaker raw equity

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Bot Architecture: Design Principles

What the Lookup Bot needs to know:

1. Current canonical hand h

Policy:

$$\pi(h, \text{pos}, \text{ctx}) \in \{\text{fold, call, raise}\}$$

Implemented as a *lookup table* no ML, no tree traversal, no bluffing.

The Lookup Bot is the simplest non-trivial agent. All future bots are benchmarked against it. Its key property: it is **deterministic** given its input, which makes it exploitable (we return to this).

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1. Current canonical hand h
2. Current position (determines which range to use)
3. Stack sizes and pot size (for raise sizing)
4. Action history: are we facing a raise?

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Lookup Bot: Ranges and Decision Engine (Python) I

```python
import random
from pypokerengine.players import BasePokerPlayer
from pypokerengine.utils.card_utils import gen_cards, estimate_hole_card_win_rate

# ============================================================
# Canonical Hand Representation (169 classes)
# ============================================================

RANK_MAP = {
    "2": 2, "3": 3, "4": 4, "5": 5, "6": 6, "7": 7,
    "8": 8, "9": 9, "T": 10, "J": 11,
    "Q": 12, "K": 13, "A": 14
}


def canonical_key(hole_card):
    # PyPokerEngine format is "SA", "DT", etc.
    r1 = RANK_MAP[hole_card[0][1]]
    r2 = RANK_MAP[hole_card[1][1]]
```

Lookup Bot: Ranges and Decision Engine (Python) II

```python
    s1 = hole_card[0][0]
    s2 = hole_card[1][0]

    hi, lo = max(r1, r2), min(r1, r2)
    suited = (s1 == s2)

    if hi == lo:
        suited = False  # pairs ignore suit

    return (hi, lo, suited)


# ============================================================
# Preflop Ranges
# ============================================================

def build_utg_range():
    pairs = {(r, r, False) for r in [14,13,12,11,10,9,8,7]}
    suited = {
        (14,13,True),(14,12,True),(14,11,True),(14,10,True),
```

Lookup Bot: Ranges and Decision Engine (Python) III

```
        (13,12,True),(13,11,True),(12,11,True),(11,10,True),(10,9,True)
    }
    offsuit = {
        (14,13,False),(14,12,False),(14,11,False)
    }
    return pairs | suited | offsuit


def build_co_range():
    R = build_utg_range()
    extra_pairs = {(6,6,False),(5,5,False)}
    extra_suited = {
        (9,8,True),(8,7,True),(7,6,True),
        (14,9,True),(14,8,True)
    }
    return R | extra_pairs | extra_suited


def build_btn_range():
    R = build_co_range()
```

Lookup Bot: Ranges and Decision Engine (Python) IV

```python
    extra_pairs = {(4,4,False),(3,3,False),(2,2,False)}
    extra_suited = {
        (6,5,True),(5,4,True),(4,3,True),
        (13,9,True),(12,9,True)
    }
    extra_offsuit = {
        (14,10,False),(13,12,False)
    }
    return R | extra_pairs | extra_suited | extra_offsuit


def build_sb_range():
    return build_btn_range()


OPENING_RANGES = {
    "UTG": build_utg_range(),
    "CO": build_co_range(),
    "BTN": build_btn_range(),
    "SB": build_sb_range()
```

Lookup Bot: Ranges and Decision Engine (Python) V

```python
}

PREMIUM = {
    (14,14,False),
    (13,13,False),
    (12,12,False),
    (14,13,True)
}


# ============================================================
# LookupBot
# ============================================================

class LookupBot(BasePokerPlayer):

    SIMULATIONS = 8000
    RAISE_THRESHOLD = 0.60

    # ------------------------------------------------------------
```

Lookup Bot: Ranges and Decision Engine (Python) VI

```python
# Position detection
# -------------------------------------------------------

def get_position(self, round_state):
    seats = round_state['seats']
    dealer_btn = round_state['dealer_btn']
    my_index = next(i for i, s in enumerate(seats)
                    if s['uuid'] == self.uuid)

    relative = (my_index - dealer_btn) % len(seats)

    if relative == 0:
        return "BTN"
    elif relative == 1:
        return "SB"
    elif relative == 2:
        return "BB"
    elif relative == 3:
        return "UTG"
    else:
```

Lookup Bot: Ranges and Decision Engine (Python) VII

```python
            return "CO"

    # -------------------------------------------------------
    # Monte Carlo Postflop Equity
    # -------------------------------------------------------

    def estimate_equity(self, hole_card, community_card, nb_players):
        return estimate_hole_card_win_rate(
            nb_simulation=self.SIMULATIONS,
            nb_player=nb_players,
            hole_card=gen_cards(hole_card),
            community_card=gen_cards(community_card)
        )

    def calculate_pot_odds(self, call_amount, pot):
        if call_amount == 0:
            return 0
        return call_amount / float(pot + call_amount)

    # -------------------------------------------------------
```

Lookup Bot: Ranges and Decision Engine (Python) VIII

```python
    # Main Decision Engine
    # -------------------------------------------------------

    def declare_action(self, valid_actions, hole_card, round_state):

        street = round_state['street']
        pot = round_state['pot']['main']['amount']
        nb_players = len(round_state['seats'])

        call_action = next(a for a in valid_actions if a['action'] == 'call')
        call_amount = call_action['amount']

        raise_action = next(a for a in valid_actions if a['action'] == 'raise')

        # ====================================================
        # PREFLOP LOOKUP ENGINE
        # ====================================================

        if street == "preflop":
```

Lookup Bot: Ranges and Decision Engine (Python) IX

```python
        position = self.get_position(round_state)
        key = canonical_key(hole_card)
        my_range = OPENING_RANGES.get(position, set())

        facing_raise = call_amount > 0

        # OPENING
        if not facing_raise:
            if key in my_range and raise_action['amount']['min'] != -1:
                return "raise", raise_action['amount']['min']
            return "fold", 0

        # FACING RAISE
        else:
            if key in PREMIUM and raise_action['amount']['min'] != -1:
                return "raise", raise_action['amount']['min'] * 3

            if key in my_range:
                return "call", call_amount
```

Lookup Bot: Ranges and Decision Engine (Python) X

```
            return "fold", 0

        # ==================================================
        # POSTFLOP MATH ENGINE
        # ==================================================

        community_card = round_state['community_card']

        equity = self.estimate_equity(hole_card, community_card, nb_players)
        pot_odds = self.calculate_pot_odds(call_amount, pot)

        if equity >= pot_odds:

            if equity >= self.RAISE_THRESHOLD and raise_action['amount']['min'] != -1:
                min_raise = raise_action['amount']['min']
                max_raise = raise_action['amount']['max']

                raise_amount = int(pot * equity)
                raise_amount = max(min_raise, min(raise_amount, max_raise))
```

Lookup Bot: Ranges and Decision Engine (Python) XI

```
                return "raise", raise_amount

            return "call", call_amount

        return "fold", 0

    # Required abstract methods
    def receive_game_start_message(self, game_info):
        pass

    def receive_round_start_message(self, round_count, hole_card, seats):
        pass

    def receive_street_start_message(self, street, round_state):
        pass

    def receive_game_update_message(self, action, round_state):
        pass

    def receive_round_result_message(self, winners, hand_info, round_state):
```

Lookup Bot: Ranges and Decision Engine (Python) XII

```
        pass
```

Variance: How Many Hands to Trust Your Win Rate?

Win rate estimator from N hands (R_i = result of hand i):

$$\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} R_i, \qquad R_i \sim (\mu, \sigma^2)$$

By CLT, a 95% confidence interval is:

$$\mu \in \hat{\mu} \pm \frac{1.96\,\sigma}{\sqrt{N}}$$

Required sample size to detect a $\mu = 5$ bb/100h win rate ($\sigma = 100$):

$$N_{\text{min}} = \left(\frac{1.96 \times 100}{5}\right)^2 \approx 1{,}537 \text{ hands}$$

But this assumes $\sigma = 100$. In reality $\sigma \approx 80$–120, so estimates are noisy over $< 10{,}000$ hands. So we must test our bots a few 10000+ times to be sure of any performance guarantees.

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A Bot That Beats Us

While LookupBot beats MathBot, it loses to a simple addition.

```
class FifteenBluffLookupBot(LookupBot):
BLUFF_FREQUENCY = 0.1515
def declare_action(self, valid_actions, hole_card, round_state) :
Get original LookupBot decision action, amount = super().declare_action(valid_actions, hole_card, round_state)
If original action is NOT fold → keep it if action != "fold": return action, amount
If original action IS fold → 15if random.random() < self.BLUFF_FREQUENCY :
call_action = next(aforainvalid_actionsifa[
'action'] == 'call' )
call_amount = call_action['amount']
If we can call, call instead of folding if call_amountisnotNone : return"call", call_amount
Otherwise keep fold return action, amount
```

Why a Deterministic Bot is Exploitable

The exploit. If our bot always folds hands outside its range, a smart opponent:

1. Observes that the bot folds $\approx 85\%$ of hands from BTN (range $\approx 15\%$)

$$\mathbb{E}[\text{steal}] = 1.5 \times \Pr[\text{bot folds}] > 0$$

Nash Equilibrium requirement. At equilibrium aka optimal play, no player can improve by deviating. A pure fold-outside-range strategy is exploitable thus, *not* an equilibrium and thus, not optimal.

The fix: Minimum Defence Frequency (MDF). We must call with at least $\mathrm{MDF} = P/(P+B)$ of our range to make the steal unprofitable.

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The exploit. If our bot always folds hands outside its range, a smart opponent:

1. Observes that the bot folds $\approx 85\%$ of hands from BTN (range $\approx 15\%$)
2. Raises with *any two cards* from the button
3. Wins the blinds (1.5 bb) every time the bot folds

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MDF: Full Derivation

Setup. Villain bets size B into a pot of P. Let $f = \Pr[\text{we fold}]$ and assume villain's bluff has 0% equity at showdown.

$$\mathbb{E}[\text{bluff}] = B \cdot f - B \cdot (1 - f) \cdot 1 = B(2f - 1)$$

Wait — villain *wins* P if we fold, *loses* B if we call. Correctly:

$$\mathbb{E}[\text{bluff}] = P \cdot f - B \cdot (1 - f)$$

Set $\mathbb{E}[\text{bluff}] = 0$ and solve for the break-even fold frequency:

$$Pf = B(1 - f) \implies f^* = \frac{B}{P + B}$$

So the **minimum call frequency** (MDF) is:

$$\boxed{\mathrm{MDF} = 1 - f^* = \frac{P}{P + B}}$$

If we fold more than f^* of our range, every bluff is instantly profitable regardless of cards.

MDF: Example and Graph

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Example. Villain bets \$50 into \$100 pot.

$$\mathrm{MDF} = \frac{100}{100 + 50} = \frac{100}{150} = 66.7\%$$

We must call with at least 66.7% of our range.

Interpretation. As bet size B grows relative to pot P, the villain is risking more per bluff, so we can fold *more* and it is still unprofitable. At a pot-sized bet ($B = P$, $\beta = 1$): MDF $= 50\%$.

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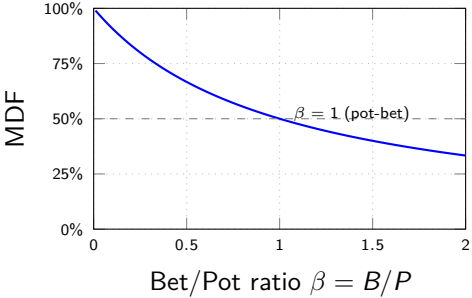

Sadly...

While the BluffBot beats the LookupBot more often than not, it still loses with significant probability.

We would like to: (1) understand *exactly* why, and (2) fix it.

Problems with the BluffBot:

- **Uniform randomness is suboptimal.** We should call/bluff with our best trash hands, not uniformly random ones. For example: a *backdoor draw* (needing both turn *and* river to complete a flush or straight, e.g. two hearts in hand with one heart on flop) has ≈3–5% equity contribution — better than pure air. We should defend with these hands first.

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- **Bet sizing is ignored.** A min-raise deserves a different response than a 3x raise.

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Questions?

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